

ARTICLE 1:

How to Refactor this Code? An Exploratory Study on Developer-ChatGPT Refactoring Conversations

2024 IEEE/ACM 21st International Conference on Mining Software Repositories (MSR)

<https://dl.acm.org/doi/10.1145/3643991.3645081>

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April 15–16, 2024

This article investigates how developers interact with ChatGPT for code refactoring tasks, analyzing over 500 real-world conversations to understand the nature, depth, and effectiveness of AI-assisted refactoring. The study finds that developers often rely on ChatGPT to identify code smells, suggest improvements, and explain refactoring rationales, highlighting the model's growing role as a collaborative coding assistant. Importantly, the research uncovers both the strengths—such as ChatGPT's ability to generate plausible and context-aware suggestions—and the limitations, including occasional superficial fixes and lack of deep architectural insight. This is significant for software development because it provides empirical evidence of how AI can support maintainability and code quality, while also emphasizing the need for developers to critically assess AI-generated suggestions, paving the way for more reliable and synergistic human-AI development workflows.

ARTICLE 2:

Student Code Refactoring Misconceptions

Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1

<https://dl.acm.org/doi/10.1145/3587102.3588840>

Eduardo Oliveira, Hieke Keuning and Johan Jeuring

2023, July 8–12

This study explores common misconceptions that students hold about code refactoring, based on an analysis of over 250 student-submitted refactoring tasks. The authors identify several prevalent misunderstandings, including the belief that refactoring should change program behavior, that renaming or reformatting alone qualifies as refactoring, and confusion between refactoring and code optimization. These misconceptions often lead students to perform unsafe or ineffective transformations, revealing gaps in conceptual understanding. The article is important for software development because it highlights the educational

challenges in teaching clean code practices and underscores the need for more effective pedagogy and tools to ensure that the next generation of developers can confidently and correctly apply refactoring techniques that improve code quality without introducing errors.

ARTICLE 3:

Artificial Intelligence Methods in Software Refactoring: A Systematic Literature Review

IEE

<https://ieeexplore.ieee.org/abstract/document/10803298>

Simona Motogna, Liviu-Marian Berciu, Vasilica-Andreea Moldovan

27 December 2024

These articles analyze studies between 2010 and 2024 on how Artificial Intelligence is applied for refactoring code. It shows positive things like the methods to use AI for refactoring but it also shows the gaps that AI can have. The articles show that detection and recommendations are the most researched refactoring stages. It also shows how the studies in AI refactoring is increasing since 2016. The authors of the article emphasize the potential of hybrid AI and LLMs (Large Language Models) to automate and improve the process of refactoring. The article also shows that the most frequently employed methods are Random Forests, Genetic Algorithms, Support Vector Machines, Convolutional Neural Networks or Decision Trees. But it also suggests that we need more research in under-explored areas like documentation.

ARTICLE 4:

Triskelion—In Pursuit of Proficiency Through Immersive Gameplay

<https://www.mdpi.com/2078-2489/16/1/28>

Victor Winter

This article talks about the effectiveness of 'Triskelion', a game that is used for learning. The article shows an experiment that was done in 2 groups. Group A practiced with the game and Group B did not have access to the game. Initial assessments showed that the baseline knowledge between the two groups was comparable. Even though it was comparable, Group A showed an improvement in the quizzes. So this outcome indicates that the immersive gameplay experience provided by the game was effective in enhancing and improving these skills for all the users, suggesting that the game was a key factor in the improvement of their level. But also, the article says that even though gamification yields positive results it is very important how you do it and you have to "know your audience".